

Arya Mohammadi

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EDUCATION

University of California San Diego

B.S. in Mathematics and Computer Science, GPA: 3.78/4.00

San Diego, CA

Expected June 2027

EXPERIENCE

Computer Science and Mathematics Tutor

Jul 2024 – Jun 2025

University of California, Riverside

Riverside, CA

- Debugged C++ codebases for 400+ students, identifying logic errors, memory issues, recursion bugs, and performance bottlenecks.
- Reviewed code for correctness, structure, and maintainability using clean code and modular design principles.
- Taught data structures, algorithms, recursion, and time complexity through hands-on problem solving and workshops.
- Led sessions on memory management and debugging workflows similar to real engineering practices.

Hardware Intern

Jun 2021 – Sep 2023

ProCustom Group

California, USA

- Executed custom hardware modifications for corporate and government clients, ensuring compliance with security standards through system integration, configuration management, and quality assurance protocols.
- Diagnosed and repaired system failures using debugging methodologies and troubleshooting techniques, replacing components and conducting performance testing and validation to meet technical specifications.

PROJECTS

Arya Electric LLC | *Next.js, Node.js, React, JavaScript, REST APIs, Playwright*

Jan 2025 – Present

- Built a full-stack platform with Next.js and RESTful API routes that processes dozens of quote requests per week, with server-side validation, sanitization, and reliable error handling.
- Developed backend systems for automated follow-ups, SMS and email workflows, webhook processing, and lead scoring, integrating Nodemailer, Twilio, and Google Places to reduce manual processing time by ~50%.
- Improved system reliability with response caching, structured error handling, and end-to-end testing (Playwright), supporting stable deployments and API response times under 500 ms for most endpoints.

DormDuos.com | *React, Node.js, MongoDB, JWT, Vercel, Railway*

Jan 2024 – Sep 2024

- Developed a full stack housing platform with a Next.js/React frontend and an Express + MongoDB REST API, supporting secure listing management, full-text search, filtering, and pagination.
- Engineered backend authentication and reliability systems including JWT auth, bcrypt hashing, sanitization middleware, retry logic, health checks, and structured error handling, blocking 100% of MongoDB injection attempts and improving query performance with indexing.
- Built comprehensive backend test coverage with Jest, Supertest, and MongoDB Memory Server, achieving 82 passing tests validating authentication, validation paths, and end-to-end API workflows.

Battlestar RPG | *C++17, STL, CMake, Make, GoogleTest*

Sep 2024 – Dec 2024

- Built core engine components including a turn-based combat engine, speed-based MaxHeap scheduler, inventory system, map navigation, and a state-driven game loop coordinating all modules.
- Designed modular architecture applying SOLID principles; implemented polymorphic item hierarchy, strategy-pattern sorting system, and template-based save/load subsystem for serializing full game state.
- Developed and validated engine functionality using GoogleTest, covering combat logic, sorting algorithms, inventory behavior, navigation rules, difficulty multipliers, and serialization workflows.

TECHNICAL SKILLS

Languages: C++, JavaScript, Python, SQL

Frameworks & Libraries: Next.js, React, Node.js, Express.js, Mongoose, Jest, Supertest, GoogleTest

Tools & Platforms: Git, GitHub, Linux, CMake, Make, MongoDB Atlas, Vercel, Railway

Technical Skills: Full-Stack Development, REST API Design, Object-Oriented Architecture, SOLID Principles, Custom Data Structures, JWT Authentication, Input Validation, Database Modeling, Indexing, Testing (Unit & Integration)